Reachability and Pareto selection with longer network trades

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Abstract

We classify improving cyclic trades on an undirected social network. For every fixed length bound $k \geq 3$, for an input bound $k \geq 3$, and for unrestricted length, both allocation reachability and agent–house reachability are NP-complete. Selecting an allocation that is Pareto optimal within the entire reachable set is polynomial-time Turing equivalent to an NP-complete decision oracle. A private-house invariant prevents longer trades from invalidating the completion reduction.

1 Model and result

There are n agents, n houses, an undirected graph on the agents, and an initial bijection. Write $M(i)$ for agent i's house in an allocation M. Each agent strictly orders her acceptable houses, which include her endowment. A trade uses distinct agents $a_1, \ldots, a_\ell$, with $2 \leq \ell \leq k$, consecutive graph edges including the closing edge, and gives a_i the current house of a_{i+1}, cyclically. Every participant must strictly improve. A two-agent trade uses one graph edge. Unrestricted length means $k = n$; an input bound is effectively capped at n.

Let $\mathcal{R}$ be all endpoints reachable from the initial allocation. A member of $\mathcal{R}$ is Pareto optimal in $\mathcal{R}$ if no other member weakly improves every agent and strictly improves at least one. The comparison is with all of $\mathcal{R}$, not just the continuations of the selected endpoint. These are the network-edge conventions in catalogue problem C78-268. The motivating survey proposes longer-cycle exchanges as a research direction without specifying these conventions [2, §3.1.3].

Theorem 1. *The following statements hold for each fixed $k \geq 3$, for an input bound $k \geq 3$, and for unrestricted length.*

1. *Reachability of a specified allocation is NP-complete.*
2. *Reachability of a specified house for a specified agent is NP-complete, even on trees.*
3. *Reachable Pareto selection admits an $\mathsf{FP}^{\mathsf{NP}}$ algorithm. Conversely, every $\mathsf{FP}^{\mathsf{NP}}$ function can be computed in polynomial time with an oracle returning any valid reachable Pareto allocation.*

The decision reductions are polynomial-time many-one reductions. The search equivalence uses polynomial-time Turing reductions that succeed for every valid oracle answer. In particular, each of the three tasks has a polynomial-time algorithm if and only if $\mathsf{P} = \mathsf{NP}$.

2 Short paths and upper bounds

Give agent i integer ranks $r_i(h) \in \{0, \ldots, n-1\}$ increasing with preference. Every trade increases $\sum_i r_i(M(i))$ by at least two, while this sum is at most $n(n-1)$. Every legal sequence therefore has at most $\lfloor n(n-1)/2 \rfloor$ trades. A certificate lists these cycles; distinctness, graph edges,

acceptability, improvement and the endpoint can all be checked in polynomial time. This proves both NP upper bounds in all three regimes.

For Pareto selection, fix agents in order. For the current agent, test her houses from best to worst, asking whether a reachable endpoint assigns that house and all previously fixed houses. Each question is in NP by the path bound. At least one answer is positive, since the previous choices preserved a reachable endpoint. After at most n^2 queries, all agents' houses are fixed. If a reachable allocation Pareto dominated the result, its first different agent would have received a better feasible choice at her selection step. This is a contradiction. All queries start from the original endowment. A witnessing path can also be recovered by polynomially many certificate-prefix queries.

3 A completion gadget that blocks mixed cycles

We use the following established hardness result: under improving *swaps*, agent–house reachability is NP-complete on trees [1, Theorem 1]. Those hard instances have a target house initially held by a different agent. Relabel the target agent as 1 and write the original house set as $X = \{x_1, \dots, x_n\}$ and the target as $t \neq x_1$. A tree has no simple cycle of length at least three, so this theorem already proves part 2 of Theorem 1.

To reduce this tree problem to allocation reachability, keep its agents and edges. For each original agent j, add a copy c_j endowed with a new house d_j, and add the edge $\{j, c_j\}$. Make the copies a clique; there are no other edges between the two parts. Original agent j ranks d_j first, then all original houses in her original order, then all other private houses. These lists may be completed strictly; all original instances used here already have complete lists.

Copy c_1 ranks t first, then d_1, then all other houses. List $X \setminus \{t\}$ as distinct houses $t_2, \dots, t_n$ in any fixed order. For $j = 2, \dots, n$, copy c_j ranks t_j first, then the other houses in the list $t_n, t_{n-1}, \dots, t_2$, then d_j, then t and all foreign private houses. Order any otherwise unspecified houses arbitrarily. The distinct top choices define one allocation

$$T(j) = d_j \quad (1 \leq j \leq n), \qquad T(c_1) = t, \qquad T(c_j) = t_j \quad (2 \leq j \leq n).$$

The construction has $2n$ agents and polynomial encoding size. It modifies the copy-clique completion used for swaps in [1, Theorem 2]; the next lemma is needed because longer trades are now allowed.

Lemma 1. *Every legal trade of length at least three in the constructed instance uses only copies. A trade between the two parts is necessarily the swap of j with c_j, after which j never trades again.*

Proof. Every agent other than j and c_j ranks d_j below her endowment. Strict improvement keeps every agent's holding at least as good as her endowment. Thus d_j can leave c_j only for j. On receiving d_j, agent j obtains her top choice and never trades again. In particular, whenever j participates in a trade, c_j still holds d_j.

Consider a legal cycle of length at least three that meets both parts. It must transfer a house from an original agent j to a copy, which can only be c_j. This copy still holds d_j and must give it to its other neighbor in the cycle, a different copy: returning it to j would repeat a vertex and make the cycle a two-agent swap. That recipient cannot improve by receiving d_j, a contradiction. No mixed cycle is legal. The original part is a tree, so it contains no remaining cycle of length at least three. Finally, a crossing swap gives j the house d_j and hence retires j permanently. □

Lemma 2. *The original tree instance can deliver t to agent 1 if and only if T is reachable in the constructed instance, for every allowed length bound at least two.*

Proof. Suppose first that original swaps deliver t to 1. Perform the same swaps in the original part, then swap each j with c_j. Original agents obtain their top choices. Copy c_1 obtains t; every other copy receives an original house different from t and therefore improves from her private house.

The remaining copies hold a permutation of $t_2, \dots, t_n$. Process $c_2, \dots, c_n$ in this order. If c_j does not hold t_j, swap it with its holder c_s. Previously processed copies already hold their distinct targets, so $s > j$. The house currently at c_j is some t_q with $q > j$. Copy c_j receives its top choice. Copy c_s prefers t_q to t_j: either $q = s$ gives its own top choice, or both houses are ordered by the common descending index order. The swap is legal. At the end the allocation is T. Only swaps were used.

Conversely, suppose a sequence reaches T. A copy other than c_1 never holds t, since she ranks it below her initial house. The first transfer of t into the copy part must therefore deliver it to c_1. By Lemma 1, this is the swap of 1 with c_1, so agent 1 held t just before that swap.

Retain only the original–original swaps preceding this event. They form a legal sequence in the original tree. To see this inductively, ignore each copy-only trade and each crossing swap. A crossing swap retires its original participant, who never appears in a later retained swap; in the projected sequence she simply keeps the original house she surrendered. Every still-active original agent has exactly the same house in the projected and enlarged sequences. Hence each retained swap has the same two houses and remains improving. The projected sequence therefore delivers t to 1. □

All newly allowed longer cycles are covered by Lemma 1, even when the bound is input or unrestricted. Lemma 2 and the source tree hardness prove part 1 of Theorem 1.

4 Pareto search hardness

Every agent has a distinct top house in the constructed instance. If the source instance is positive, T is reachable and Pareto dominates every different allocation, so every valid reachable Pareto answer is T. If the source instance is negative, T is unreachable, so every valid answer differs from T. One oracle call and an equality test therefore decide the NP-complete tree problem. This is the all-top argument for swaps in [1, Proposition 4], with longer-cycle soundness supplied by Lemma 1.

More precisely, let f be computed by a polynomial-time transducer with an NP oracle. Reduce each oracle question to the NP-complete tree problem, construct the gadget, call a reachable Pareto oracle and test whether its answer is T. The answer is correct regardless of which valid Pareto allocation the oracle returns. Replacing every adaptive NP query in this way computes f in polynomial time. Together with the selector already proved, this establishes part 3 in the stated Turing sense. It does not assert polynomial-time verification of arbitrary Pareto answers or completeness under a stronger reduction notion.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

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